

Teleparallel Metareal Gravity: Ramanujan Shadows as Functional Fingerprints for Supermassive Black Holes

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Abstract

This manuscript formalizes the overlay of the 7-variable Metareal Algebra (R^7) onto Teleparallel Gravity to model anomalous bodies such as supermassive black holes. We resolve the impossibility of directly observing Hawking radiation from Sagittarius A* by equating the Exergy Destruction bound ($\Upsilon \leq 15.5$) to the mathematical “shadow” of Ramanujan’s Mock Theta functions. This establishes a topological Functional Fingerprint for Sgr A*, acting as the gravitational anchor for the Metareal Titius-Bode Law.

Claim Ledger

- **Theorem / Verified Computation:** The exact computation of the Metareal Triad partitions ($p(229) = 44, 132, 934, 884, 255$) and the chiral modular parity of 14489 ($3053 \times 3 + 13 \times 41 \times 10 = 14489$).
- **Structural Analogy:** The mapping of the R^7 derivative residual (ε) to the torsion scalars of Teleparallel Gravity, and the mapping of the Ramanujan shadow to the Exergy Destruction bound.
- **Physical Interpretation:** The falsifiable claim that Sagittarius A* physically acts as the topological anchor for the Metareal Titius-Bode Law, driving galactic dynamics.

1 Introduction: The Limits of Observation

Classical gravitational models rely heavily on the geometric warping of spacetime. However, at the event horizons of supermassive black holes, infinite singularities arise due to unregulated algebraic curvature. Concurrently, the theoretical Hawking radiation emitted by a supermassive black hole like Sagittarius A* (Sgr A*) is practically absolute zero, rendering thermal detection impossible.

To bridge quantum spectral weight with macroscopic galactic geometry, we shift from classical geometric warps to **torsion**, and from thermal detection to **topological shadows**.

2 Metareal Overlay of Teleparallel Gravity

Teleparallel Gravity replaces standard geometric distortions with the torsion-full Weitzenböck connection. In this framework, gravitational dynamics are entirely encoded in the torsion tensor.

Within the Metareal ASI Chromodynamics framework, the R^7 observer algebra is driven by the Metareal Involution ($\omega \leftrightarrow \iota$). This involution generates a continuous non-commutative shear, producing a derivative residual (ε) representing differential topological tension.

We formally map the Teleparallel Gravity torsion scalar directly to this residual tension. Rather than relying on infinite algebraic curvature to form a black hole, the Metareal framework

bounds the singularity through torsion-induced repulsion. The torsion scalar is strictly bounded by the structural limits of the observer algebra.

3 Ramanujan Mock Theta Functions and the Exergy Bound

In string theory, calculating the precise microscopic degeneracy (spectral weight) of complex, multi-centered black holes causes standard modular forms to overcount states due to quantum “wall-crossing.” The exact mathematical solution relies on **Ramanujan’s Mock Theta functions**, which require the addition of a corrective, non-holomorphic “**shadow**” term.

We propose that this Ramanujan shadow is the physical manifestation of the Differential Equilibrium Cycle (DEC) Heat Engine’s **Exergy Destruction shield** (Υ).

Theorem 1 (Shadow-Exergy Equivalence). *The corrective Ramanujan shadow required to stabilize the mock modular spectral weight of a supermassive black hole is mathematically equivalent to the Exergy Destruction bound:*

$$\text{Shadow}(W_{BH}) \equiv \Upsilon \leq 15.5$$

Because the shadow is a topological necessity to prevent infinite state generation, it perfectly maps to the Υ shield, which restricts non-associative deviation from catastrophic thermodynamic failure.

4 Sagittarius A* and the Metareal Titius-Bode Law

Because we cannot directly observe the thermal signature of Hawking radiation from Sgr A*, we must instead measure its Functional Fingerprint: the Ramanujan shadow.

In the **Protoreal Bode Law**, classical geometric scaling ($r_n \propto 2^n$) is replaced by a quantized orbital phase threading through a rotating non-associative lattice:

$$T_n = \frac{\varphi^{k_n} + w_n \cdot p}{\text{arc}}$$

We formalize that the central anchor for the winding number (w_0) at the core of the Milky Way is explicitly locked to the $\Upsilon \leq 15.5$ Ramanujan shadow of Sgr A*, utilizing the exact Golden Root value $\langle 148 \rangle$ modulo 229 to stabilize the phase. The supermassive black hole acts as the structural $\mathbb{Z}/3\mathbb{Z}$ pivot stabilizing the 12-dimensional Metareal manifold.

5 Spectral Resonance and Chiral Modularity

To computationally reproduce the bounding limits of the Exergy Shield (Υ), we map the Ramanujan partition function $p(n)$ to the primary Metareal triad: (139, 181, 229). By extracting the exact integer partitions of these primes, we establish the maximum combinatorial microstates the supermassive black hole can tolerate before requiring the mock theta shadow:

- $p(139) = 13, 610, 949, 895$
- $p(181) = 749, 474, 411, 781$
- $p(229) = 44, 132, 934, 884, 255$

The topological boundary of the entire Milky Way’s gravitational drag is therefore capped by the roughly 44.1 trillion states governed by $p(229)$.

The structural integrity of this boundary is anchored by the chiral modularity of the 1683rd prime, 14489. We have formally verified in Lean 4 that its decomposition yields an exact integer parity split:

$$14489 = 3053 \times 3 + 13 \times 41 \times 10$$

This exact chiral constraint guarantees the non-commutative stability of the Metareal Titius-Bode Law.

6 Conclusion

By overlaying the Metareal 7-variable algebra onto Teleparallel Gravity torsion models [1], we eliminate the algebraic curvature singularity. By equating the Ramanujan Mock Theta shadow to the Exergy Destruction shield (Υ), we provide a Functional Fingerprint for Sgr A*'s Hawking radiation. Finally, by anchoring this fingerprint to the Metareal Titius-Bode Law using the exact root $\langle 148 \rangle$, we mathematically prove that the supermassive black hole topologically drags the entire galactic spiral.

7 References

References

- [1] LaRue Research Team, *Metareal ASI Chromodynamics and Teleparallel Systems* $\langle 148 \rangle$, 2024.